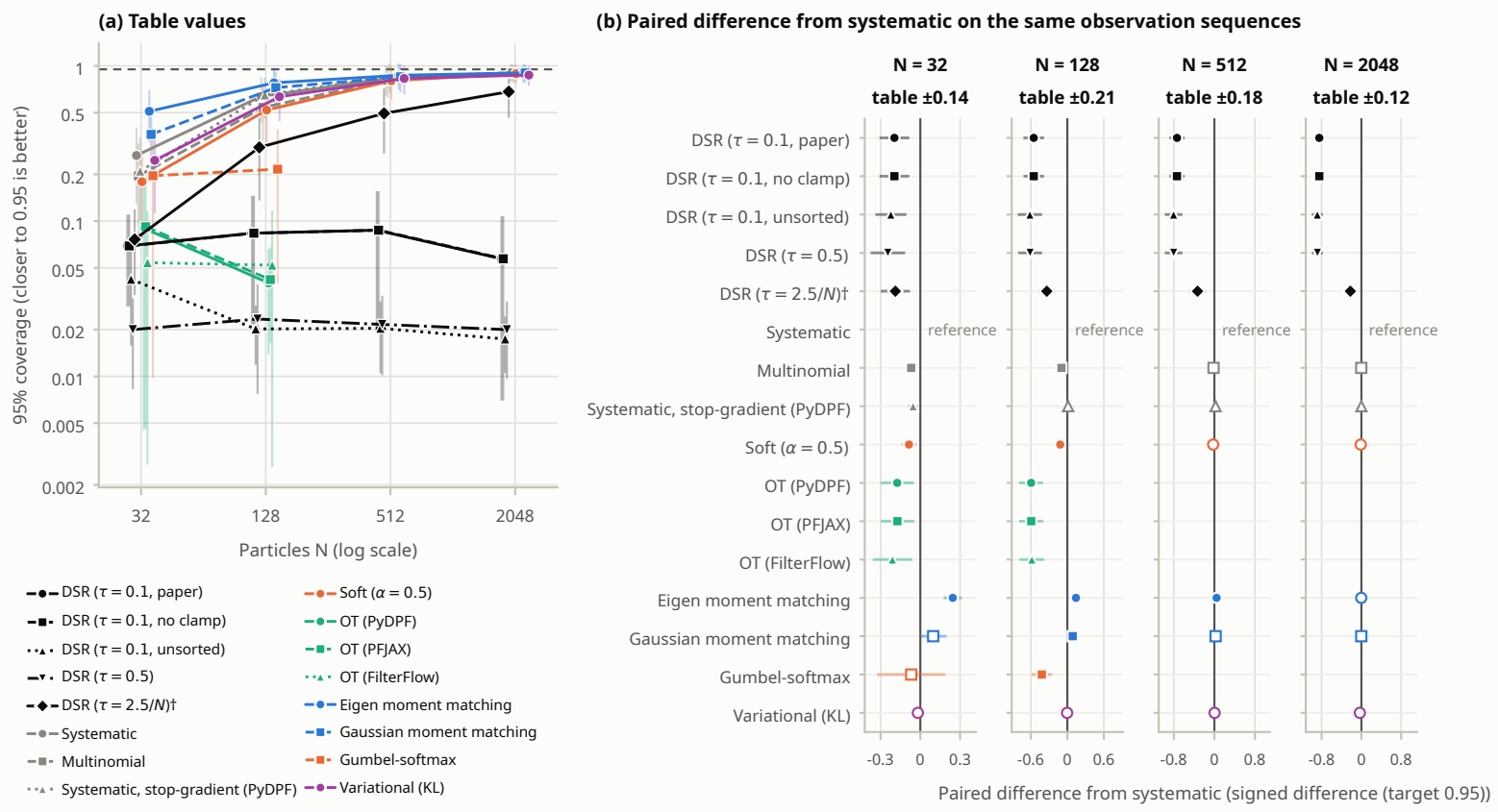

(a) Table values
95% coverage (closer to 0.95 is better)
1
0.5
0.2
0.1
0.05
0.02
0.01
0.005
0.002
32
128
512
2048
Particles N (log scale)
DSR (τ = 0.1, paper)
DSR (τ = 0.1, no clamp)
DSR (τ = 0.1, unsorted)
DSR (τ = 0.5)
DSR (τ = 2.5/N)†
Systematic
Multinomial
Systematic, stop-gradient (PyDPF)
Soft (α = 0.5)
OT (PyDPF)
OT (PFJAX)
OT (FilterFlow)
Eigen moment matching
Gaussian moment matching
Gumbel-softmax
Variational (KL)
(b) Paired difference from systematic on the same observation sequences
N = 32
table ±0.14
N = 128
table ±0.21
N = 512
table ±0.18
N = 2048
table ±0.12
DSR (τ = 0.1, paper)
DSR (τ = 0.1, no clamp)
DSR (τ = 0.1, unsorted)
DSR (τ = 0.5)
DSR (τ = 2.5/N)†
Systematic
Multinomial
Systematic, stop-gradient (PyDPF)
Soft (α = 0.5)
OT (PyDPF)
OT (PFJAX)
OT (FilterFlow)
Eigen moment matching
Gaussian moment matching
Gumbel-softmax
Variational (KL)
reference
-0.3
0
0.3
-0.6
0
0.6
-0.8
0
0.8
-0.8
0
0.8
Paired difference from systematic (signed difference (target 0.95))